

Linear Algebra MID

Show equations, substitutions, intermediate reasoning, and a final labeled answer.

1. [8 pts] Solve $2x+4y=10$ and $2x+3y=8$. Show elimination and verify by substitution.

2. [6 pts] For $A=\begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix}$, compute $\det(A)$ and state whether A is invertible. Justify the conclusion.

3. [8 pts] Let $T(x,y)=(x+2y, 3x-y)$. Find $T(1,-1)$, write the matrix of T , and state whether T is linear.

4. [6 pts] Give a test for linear independence of vectors and apply it to $\{(1,0),(0,1)\}$.

5. [5 pts] Explain geometrically what the null space of a matrix represents and how it relates to $Ax=0$.

6. [8 pts] Solve $5x+2y=9$ and $2x+5y=12$. Show elimination and verify by substitution.

7. [6 pts] For $A=\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$, compute $\det(A)$ and state whether A is invertible. Justify the conclusion.

8. [8 pts] Let $T(x,y)=(x+2y, 3x-y)$. Find $T(1,-1)$, write the matrix of T , and state whether T is linear.
